

MATH 4820 / Regression Analysis

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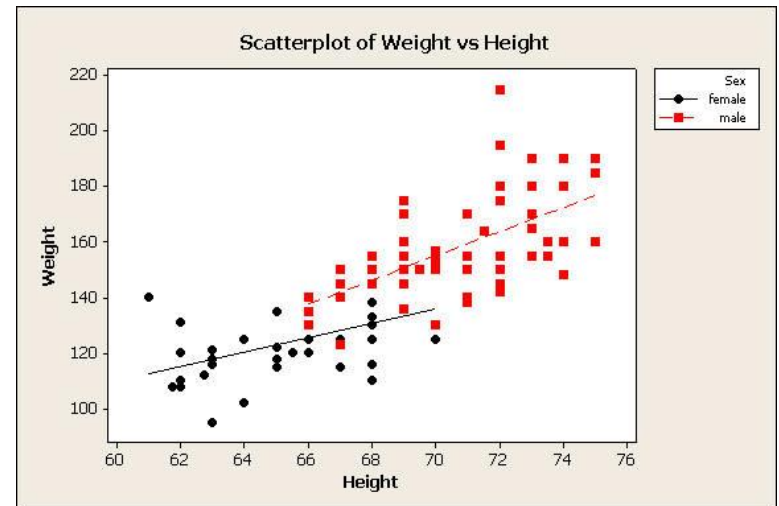
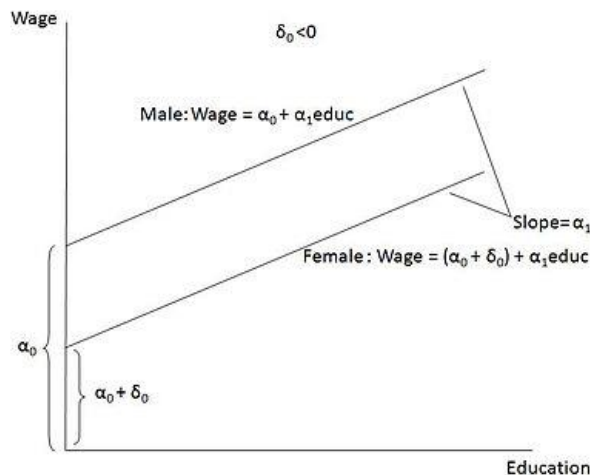
**Indicator
Variables**



Department of Mathematics, Statistics and Computer Science

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

- **Qualitative variables** – also known as categorical variables. Qualitative variables do not have a scale of measurement.
- **Indicator variables** – a variable that assigns levels to the qualitative variable (also known as **dummy variables**).



8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

Example

- Relate the effective life of a cutting tool (y) used on a lathe to the lathe speed in revolutions per minute (x_1) and type of cutting tool used (x_2).
- Tool type is qualitative and can be represented as:

$$x_2 = \begin{cases} 0 & \text{Tool A} \\ 1 & \text{Tool B} \end{cases}$$

- If a first-order model is appropriate:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

Example

For Tool type A this model becomes:

$$y = \beta_0 + \beta_1 x_1 + \varepsilon$$

For Tool type B this model becomes:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 + \varepsilon$$

$$y = (\beta_0 + \beta_2) + \beta_1 x_1 + \varepsilon$$

- Changing from A to B induces a *change* in the intercept (slope is unchanged and identical). We assume that the variance is equal for all levels of the qualitative variable.

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

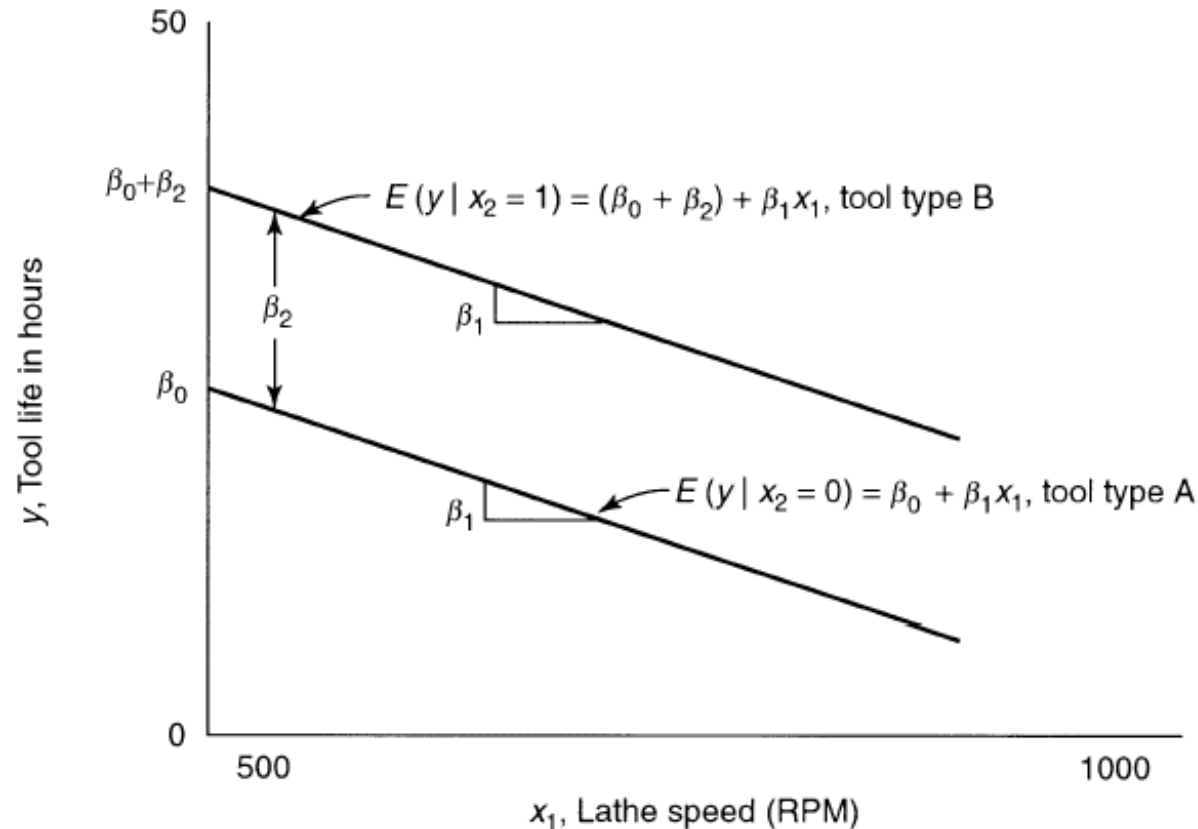


Figure 8.1 Response functions for the tool life example.

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

- For qualitative variables with a levels, we would need $a-1$ indicator variables. For example, say there were three tool types, A, B, and C. Then two indicator variables (called x_2 and x_3) will be needed:

x_2	x_3	
0	0	if the observation is from tool type A
1	0	if the observation is from tool type B
0	1	if the observation is from tool type C

the regression model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

EXAMPLE 8.1 TOOL LIFE DATA

TABLE 8.1 Data, Fitted Values, and Residuals for Example 8.1

i	y_i (Hours)	x_{i1} (rpm)	Tool Type	$\hat{y}_i$	e_i
1	18.73	610	A	20.7552	-2.0252
2	14.52	950	A	11.7087	2.8113
3	17.43	720	A	17.8284	-0.3984
4	14.54	840	A	14.6355	-0.0955
5	13.44	980	A	10.9105	2.5295
6	24.39	530	A	22.8838	1.5062
7	13.34	680	A	18.8927	-5.5527
8	22.71	540	A	22.6177	0.0923
9	12.68	890	A	13.3052	-0.6252
10	19.32	730	A	17.5623	1.7577
11	30.16	670	B	34.1630	-4.0030
12	27.09	770	B	31.5023	-4.4123
13	25.40	880	B	28.5755	-3.1755
14	26.05	1000	B	25.3826	0.6674
15	33.49	760	B	31.7684	1.7216
16	35.62	590	B	36.2916	-0.6716
17	26.07	910	B	27.7773	-1.7073
18	36.78	650	B	34.6952	2.0848
19	34.95	810	B	30.4380	4.5120
20	43.67	500	B	38.6862	4.9838

EXAMPLE 8.1 TOOL LIFE DATA

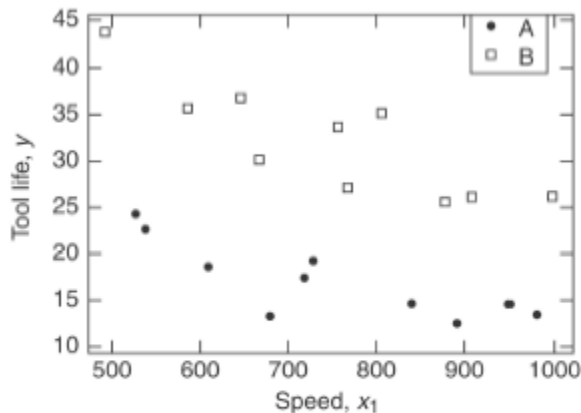


Figure 8.2 Plot of tool life y versus lathe speed x_1 for tool types A and B.

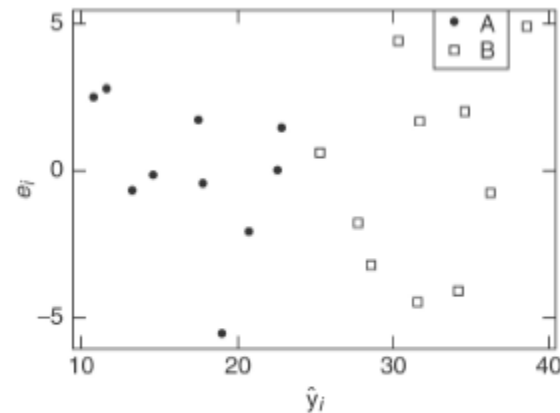


Figure 8.3 Plot of residuals e_i versus fitted values $\hat{y}_i$, Example 8.1.

- The model to be fit is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

where $x_2 = 0$ indicates Tool type A, if $x_2 = 1$ then Tool type B is used.

- The least squares fit is

$$\hat{y} = 36.986 - 0.027x_1 + 15.004x_2$$

R CODE

- `tool=read.table(file.choose(),header=T)`
- `tool.lm=lm(y~x+factor(Tool),data=tool)`

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  36.98560    3.51038   10.536 7.16e-09 ***
xil..rpm.    -0.02661    0.00452   -5.887 1.79e-05 ***
Tool.TypeB   15.00425    1.35967   11.035 3.59e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.039 on 17 degrees of freedom
Multiple R-squared:  0.9003,    Adjusted R-squared:  0.8886
F-statistic: 76.75 on 2 and 17 DF,  p-value: 3.086e-09

```

TABLE 8.2 Summary Statistics for the Regression Model in Example 8.1

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	1418.034	2	709.017	76.75	3.12×10^{-9}
Residual	157.055	17	9.239		
Total	1575.089	19			

Coefficient	Estimate	Standard Error	t_0	P Value
β_0	36.986			
β_1	-0.027	0.005	-5.887	8.97×10^{-6}
β_2	15.004	1.360	11.035	1.79×10^{-9}

$R^2 = 0.9003$

**EXAMPLE 8.1 TOOL LIFE DATA – RESIDUAL ANALYSIS – SEE
PLOT OF RESIDUALS VERSUS FITTED VALUES IN FIG. 8.3
(SLIDE 8) AND NORMAL PROBABILITY PLOT BELOW:**

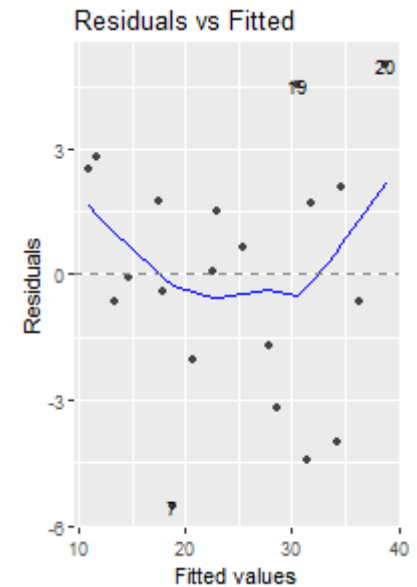
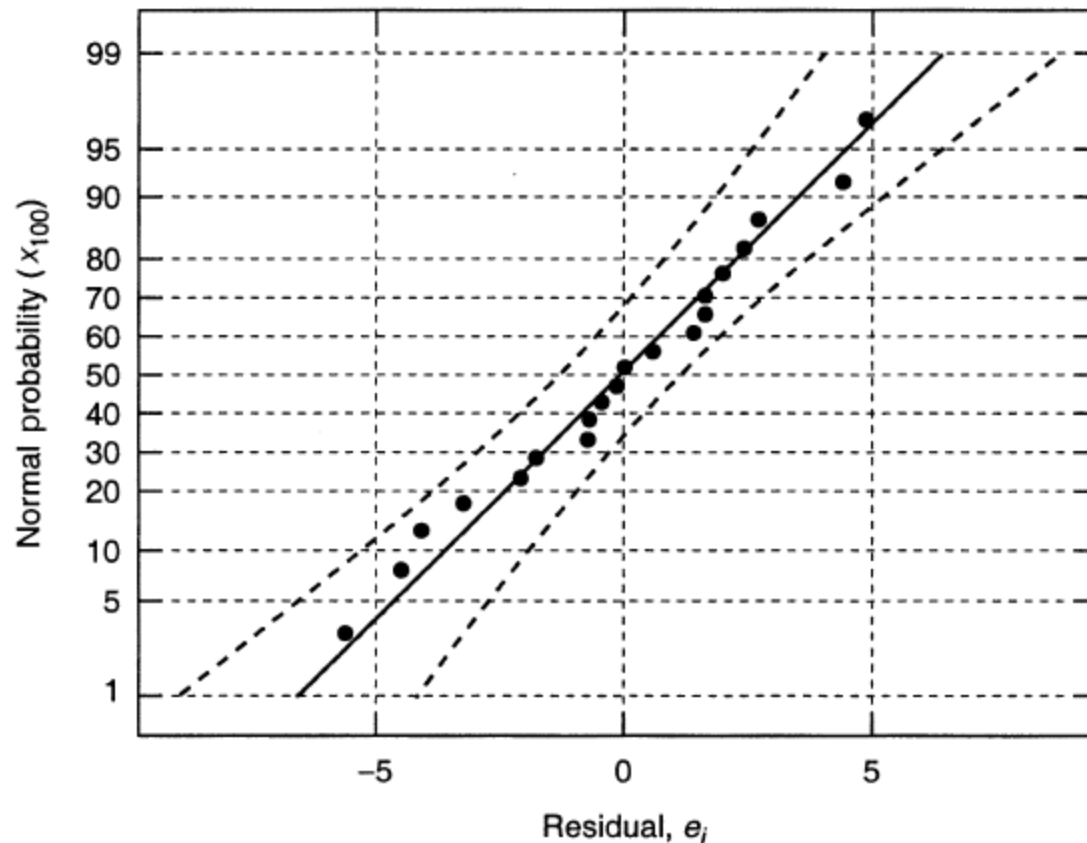


Figure 8.4 Normal probability plot of residuals, Example 8.1.

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

- Two separate models could have been fit to the data.
- However, the single-model approach is preferred because the analyst has only one final equation to work with instead of two, a much simpler practical result.
- Furthermore, since both straight lines are assumed to have the same slope, it makes sense to combine the data from both tool types to produce a single estimate of this common parameter.

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

Difference in Slope

- If we expect the slopes to differ, we can model this phenomenon by including an interaction term between the variables.
- Consider the tool life data again, and say we believe there may be different slopes for the two tools. The model we can fit to account for the change in slope is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

Difference in Slope

- If tool type A is used: $y = \beta_0 + \beta_1 x_1 + \varepsilon$
- If tool type B is used: $y = \beta_0 + \beta_1 x_1 + \beta_2(1) + \beta_3 x_1(1) + \varepsilon$
$$= (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x_1 + \varepsilon$$
- Thus, the intercept has shifted and so has the slope.
- β_2 – change in the intercept caused by changing from type A to type B
- β_3 – change in the slope caused by changing from type A to type B

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

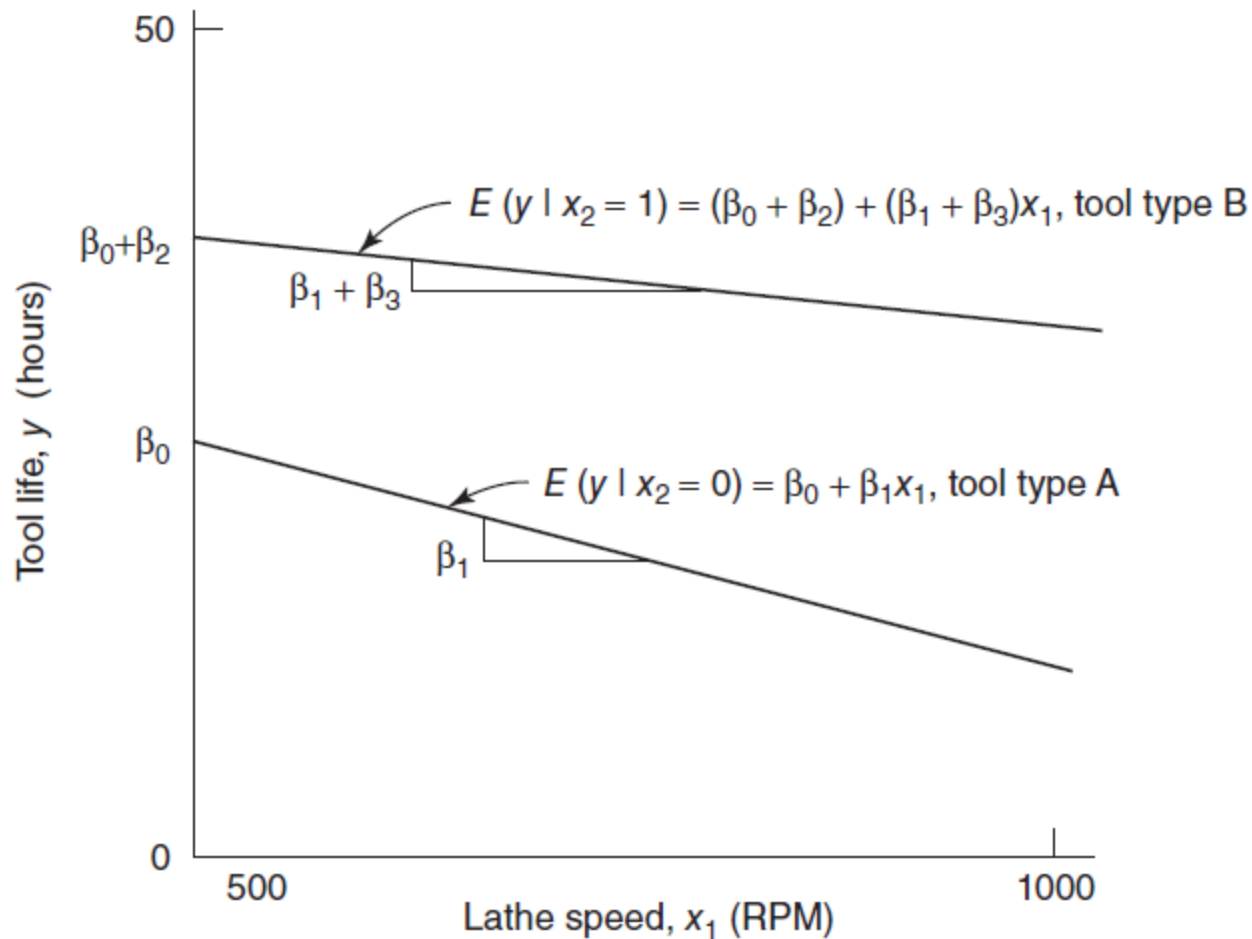


Figure 8.5 Response functions for Eq. (8.4).

8.1 THE GENERAL CONCEPT OF INDICATOR VARIABLES

- What we really have are two regression equations; one for tool type A and one for tool type B. If we wanted to test to determine if these two equations are the same, we can use the extra sum of squares method and conduct a test of hypothesis.

$$H_0: \beta_2 = \beta_3 = 0 \text{ vs. } H_1: \beta_2 \neq 0 \text{ and/or } \beta_3 \neq 0$$

- The test statistic would be

$$F_0 = \frac{SS_R(\beta_2, \beta_3 | \beta_1, \beta_0) / 2}{MS_{\text{Res}}}$$

EXAMPLE 8.2 THE TOOL LIFE DATA

We will fit the regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$

The fitted regression model is

$$\hat{y} = 32.775 - 0.021x_1 + 23.971x_2 - 0.012x_1x_2$$

TABLE 8.3 Summary Analysis for the Tool Life Regression Model in Example 8.2

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	1434.112	3	478.037	54.25	1.32×10^{-9}
Error	140.976	16	8.811		
Total	1575.088	19			

Coefficient	Estimate	Standard Error	t_0	Sum of Squares
β_0	32.775			
β_1	-0.021	0.0061	-3.45	$SS_R(\beta_1 \beta_0) = 293.005$
β_2	23.971	6.7690	3.54	$SS_R(\beta_2 \beta_1, \beta_0) = 1125.029$
β_3	-0.012	0.0088	-1.35	$SS_R(\beta_3 \beta_2, \beta_1, \beta_0) = 16.078$
$R^2 = 0.9105$				

EXAMPLE 8.3 AN INDICATOR VARIABLE WITH MORE THAN TWO LEVELS

- An electric utility is investigating the effect of the size of a single-family house (x_1) and the type of air conditioning used in the house on the total electricity consumption during warm weather months (y).

Type of Air Conditioning	x_2	x_3	x_4
No air conditioning	0	0	0
Window units	1	0	0
Heat pump	0	1	0
Central air conditioning	0	0	1

EXAMPLE 8.3 AN INDICATOR VARIABLE WITH MORE THAN TWO LEVELS

The regression model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon$$

If the house has no air conditioning, Eq. (8.7) becomes

$$y = \beta_0 + \beta_1 x_1 + \varepsilon$$

If the house has window units, then

$$y = (\beta_0 + \beta_2) + \beta_1 x_1 + \varepsilon$$

If the house has a heat pump, the regression model is

$$y = (\beta_0 + \beta_3) + \beta_1 x_1 + \varepsilon$$

while if the house has central air conditioning, then

$$y = (\beta_0 + \beta_4) + \beta_1 x_1 + \varepsilon$$

EXAMPLE 8.3 AN INDICATOR VARIABLE WITH MORE THAN TWO LEVELS

- In this problem it would seem unrealistic to assume that the slope of the regression function relating mean electricity consumption to the size of the house does not depend on the type of air conditioning system.
- For example, we would expect the mean electricity consumption to increase with the size of the house, but the rate of increase should be different for a central air conditioning system than for window units because central air conditioning should be more efficient than window units for larger houses.
- There should be an **interaction** between the size of the house and the type of air conditioning system.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_2 + \beta_6 x_1 x_3 + \beta_7 x_1 x_4 + \varepsilon$$

EXAMPLE 8.3 AN INDICATOR VARIABLE WITH MORE THAN TWO LEVELS

- The four regression models corresponding to the four types of air conditioning systems are as follows:

$$y = \beta_0 + \beta_1 x_1 + \varepsilon_1 \quad (\text{no air conditioning})$$

$$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_5)x_1 + \varepsilon \quad (\text{window units})$$

$$y = (\beta_0 + \beta_3) + (\beta_1 + \beta_6)x_1 + \varepsilon \quad (\text{heat pump})$$

$$y = (\beta_0 + \beta_4) + (\beta_1 + \beta_7)x_1 + \varepsilon \quad (\text{central air conditioning})$$

Example 8.4 More than Two Indicator Variables

- Suppose that in Example 8.1 a second qualitative factor, the type of cutting oil used, must be considered.
- Assuming that this factor has two levels, we may define a second indicator variable, x_3 , as follows:

EXAMPLE 8.4 MORE THAN TWO INDICATOR VARIABLES

$$x_3 = \begin{cases} 0 & \text{if low-viscosity oil used} \\ 1 & \text{if medium-viscosity oil used} \end{cases}$$

A regression model relating tool life (y) to cutting speed (x_1), tool type (x_2), and type of cutting oil (x_3) is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon \quad (8.9)$$

Clearly the slope β_1 of the regression model relating tool life to cutting speed does not depend on either the type of tool or the type of cutting oil. The intercept of the regression line depends on these factors in an additive fashion.

Various types of interaction effects may be added to the model. For example, suppose that we consider interactions between cutting speed and the two qualitative factors, so that model (8.9) becomes

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon \quad (8.10)$$

EXAMPLE 8.4 MORE THAN TWO INDICATOR VARIABLES

Various types of interaction effects may be added to the model. For example, suppose that we consider interactions between cutting speed and the two qualitative factors, so that model (8.9) becomes

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon \quad (8.10)$$

This implies the following situation:

Tool Type	Cutting Oil	Regression Model
A	Low viscosity	$y = \beta_0 + \beta_1 x_1 + \varepsilon$
B	Low viscosity	$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_4)x_1 + \varepsilon$
A	Medium viscosity	$y = (\beta_0 + \beta_3) + (\beta_1 + \beta_5)x_1 + \varepsilon$
B	Medium viscosity	$y = (\beta_0 + \beta_2 + \beta_3) + (\beta_1 + \beta_4 + \beta_5)x_1 + \varepsilon$

EXAMPLE 8.4 MORE THAN TWO INDICATOR VARIABLES

Suppose that we add a cross-product term involving the two indicator variables x_2 and x_3 to the model, resulting in

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \beta_6 x_2 x_3 + \varepsilon \quad (8.11)$$

We then have the following:

Tool Type	Cutting Oil	Regression Model
A	Low viscosity	$y = \beta_0 + \beta_1 x_1 + \varepsilon$
B	Low viscosity	$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_4)x_1 + \varepsilon$
A	Medium viscosity	$y = (\beta_0 + \beta_3) + (\beta_1 + \beta_5)x_1 + \varepsilon$
B	Medium viscosity	$y = (\beta_0 + \beta_2 + \beta_3 + \beta_6) + (\beta_1 + \beta_4 + \beta_5)x_1 + \varepsilon$

8.2 SOME COMMENTS ON INDICATOR VARIABLES

- Try to avoid using a specific metric for the levels of the qualitative variable (avoid allocating codes). That is, it is dangerous to assign values to each level such as 1, 2, 3, and 4. Why?
- You can substitute indicator variables for quantitative regressors.
 - Useful if accurate data cannot be readily attained.
 - Group the data into classes or intervals and then assign indicator variables.
 - Drawback: loss of information by not using the actual data. Quantitative information is often more useful than qualitative.

8.3 REGRESSION APPROACH TO ANALYSIS OF VARIANCE

- The **analysis of variance** is a technique frequently used to analyze data from **planned** or **designed experiments**.
- Essentially, any analysis-of-variance problem can be treated as a regression problem in which all of the regressors are indicator variables.

The model for the one-way classification analysis of variance is

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, \dots, k, \quad j = 1, 2, \dots, n \quad (8.15)$$

where y_{ij} is the j th observation for the i th **treatment** or **factor level**, μ is a parameter common to all k treatments (usually called the **grand mean**), τ_i is a parameter that represents the effect of the i th treatment, and ε_{ij} is an $\text{NID}(0, \sigma^2)$ error component. It is customary to define the treatment effects in the balanced case (i.e., an equal number of observations per treatment) as

$$\tau_1 + \tau_2 + \dots + \tau_k = 0$$

8.3 REGRESSION APPROACH TO ANALYSIS OF VARIANCE

- In the fixed-effects or model I case, the analysis of variance is used to test the hypothesis that all k population means are equal, or equivalently,

$$H_0: \tau_1 = \tau_2 = \cdots = \tau_k = 0$$

$$H_1: \tau_i \neq 0 \quad \text{for at least one } i$$

TABLE 8.4 One-Way Analysis of Variance

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Treatments	$n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2$	$k - 1$	$\frac{SS_{\text{Treatments}}}{k - 1}$	$\frac{MS_{\text{Treatments}}}{MS_{\text{Res}}}$
Error	$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$	$k(n - 1)$	$\frac{SS_{\text{Res}}}{k(n - 1)}$	
Total	$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$	$kn - 1$		

8.3 REGRESSION APPROACH TO ANALYSIS OF VARIANCE

To illustrate the connection between the single-factor fixed-effects analysis of variance and regression, suppose that we have $k = 3$ treatments, so that Eq. (8.15) becomes

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, 3, \dots, n \quad (8.17)$$

These three treatments may be viewed as three levels of a **qualitative factor**, and they can be handled using indicator variables. Specifically a qualitative factor with three levels would require two indicator variables defined as follows:

$$x_1 = \begin{cases} 1 & \text{if the observation is from treatment 1} \\ 0 & \text{otherwise} \end{cases}$$

$$x_2 = \begin{cases} 1 & \text{if the observation is from treatment 2} \\ 0 & \text{otherwise} \end{cases}$$

8.3 REGRESSION APPROACH TO ANALYSIS OF VARIANCE

Therefore, the regression model becomes

$$y_{ij} = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, 2, \dots, n \quad (8.18)$$

The regression model (8.18) becomes

$$\begin{aligned} y_{1j} &= \beta_0 + \beta_1(1) + \beta_2(0) + \varepsilon_{1j} \\ &= \beta_0 + \beta_1 + \varepsilon_{1j} \end{aligned}$$

In general, if there are k treatments, the regression model for the single-factor analysis of variance will require $k - 1$ indicator variables, for example,

$$\begin{aligned} y_{ij} &= \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \dots + \beta_{k-1} x_{k-1,j} + \varepsilon_{ij} \\ i &= 1, 2, \dots, k, \quad j = 1, 2, \dots, n \quad (8.19) \end{aligned}$$

where

$$x_{ij} = \begin{cases} 1 & \text{if observation } j \text{ is from treatment } i \\ 0 & \text{otherwise} \end{cases}$$

QUESTIONS?

- ANY QUESTION?

EXAMPLE 8.5 COMPARING REGRESSION MODELS

Consider the case of simple linear regression where the n observations can be formed into M groups, with the m th group having n_m observations. The most general model consists of M separate equations, such as

$$y = \beta_{0m} + \beta_{1m}x + \varepsilon, \quad m = 1, 2, \dots, M \quad (8.12)$$

It is often of interest to compare this general model to a more restrictive one.

EXAMPLE 8.5 COMPARING REGRESSION MODELS

a. Parallel Lines In this situation all M slopes are identical, $\beta_{11} = \beta_{12} = \dots = \beta_{1M}$, but the intercepts may differ. Note that this is the type of problem encountered in Example 8.1 (where $M = 2$), leading to the use of an additive indicator variable. More generally we may use the extra-sum-of squares method to test the hypothesis $H_0: \beta_{11} = \beta_{12} = \dots = \beta_{1M}$. Recall that this procedure involves fitting a **full model** (FM) and a **reduced model** (RM) restricted to the null hypothesis and computing the F statistic:

$$F_0 = \frac{[SS_{\text{Res}}(RM) - SS_{\text{Res}}(FM)] / (df_{RM} - df_{FM})}{SS_{\text{Res}}(FM) / df_{FM}} \quad (8.13)$$

If the reduced model is as satisfactory as the full model, then F_0 will be small compared to $F_{\alpha, df_{RM} - df_{FM}, df_{FM}}$. Large values of F_0 imply that the reduced model is inadequate.

EXAMPLE 8.5 COMPARING REGRESSION MODELS

To fit the full model (8.12), simply fit M separate regression equations. Then $SS_{\text{Res}}(FM)$ is found by adding the residual sums of squares from each separate regression. The degrees of freedom for $SS_{\text{Res}}(FM)$ is $df_{FM} = \sum_{m=1}^M (n_m - 2) = n - 2M$. To fit the reduced model, define $M - 1$ indicator variables $D_1, D_2, \dots, D_{M-1}$ corresponding to the M groups and fit

$$y = \beta_0 + \beta_1 x + \beta_2 D_1 + \beta_3 D_2 + \dots + \beta_M D_{M-1} + \varepsilon$$

The residual sum of squares from this model is $SS_{\text{Res}}(RM)$ with $df_{RM} = n - (M + 1)$ degrees of freedom.

EXAMPLE 8.5 COMPARING REGRESSION MODELS

b. Concurrent Lines In this section, all M intercepts are equal, $\beta_{01} = \beta_{02} = \dots = \beta_{0M}$, but the slopes may differ. The reduced model is

$$y = \beta_0 + \beta_1 x + \beta_2 Z_1 + \beta_2 Z_2 + \dots + \beta_M Z_{M-1} + \varepsilon$$

where $Z_k = xD_k$, $k = 1, 2, \dots, M - 1$. The residual sum of squares from this model is $SS_{\text{Res}}(RM)$ with $df_{RM} = n - (M + 1)$ degrees of freedom. Note that we are assuming concurrence at the origin. The more general case of concurrence at an arbitrary point x_0 is treated by Graybill [1976] and Seber [1977].

EXAMPLE 8.5 COMPARING REGRESSION MODELS

c. Coincident Lines In this case both the M slopes and the M intercepts are the same, $\beta_{01} = \beta_{02} = \cdots = \beta_{0M}$, and $\beta_{11} = \beta_{12} = \cdots = \beta_{1M}$. The reduced model is simply

$$y = \beta_0 + \beta_1 x + \varepsilon$$

and the residual sum of squares $SS_{\text{Res}}(RM)$ has $df_{RM} = n - 2$ degrees of freedom. Indicator variables are not necessary in the test of coincidence, but we include this case for completeness.

QUESTIONS?

- ANY QUESTION?